

BIOGRAPHICAL SKETCH

Name: **Tapan Kumar Nayak**

Designation: Assistant Professor

Department/Institution: **Kusuma School of Biological Sciences, IIT Delhi, New Delhi**

Date of Birth: 24/10/1980

Sex: Male

SC/ST: N.A.

Education: (Post graduation onwards and professional career)

Sl. No	Institution Place	Degree Awarded	Year	Field of Study
1	IISc Bangalore	Ph.D.	2010	Biophysics
2	JNU, New Delhi	MSc.	2004	Life Sciences

A. Position and Honors

Position and Employment (starting with most recent employer)

Sl. No	Institution Place	Position	From	To
1	IIT Delhi, New Delhi	Asst. Professor	05/12/2018	Present
2	Nanon Technologies, New Jersey, US	Sr. Application Scientist	09/05/2018	28/11/2018
3	SUNY at Buffalo, NY	Asst. Professor (Research)	01/04/2016	08/05/2018
4	SUNY at Buffalo, NY	Postdoctoral Res. Associate	05/05/2010	31/03/2016

Honors and Awards

1. AHA Postdoctoral fellowship, American Heart Association, 18 % score. 2014
2. Young investigator award, International Union of Physiological Sciences. 2009
3. Research scholar travel grant for international symposium, CSIR-UGC, India. 2009
4. Best poster award, CMC Cardiology winter symposium, Vellore, India. 2008
5. Research Fellowship, CSIR-UGC, India. 2004
6. GATE, Life Sciences, 98.47 % score (75 rank of ~ 8000 candidates). 2004
7. University 1st rank, Botany (Honors), B.Sc., Berhampur University, India. 2002

Professional Experience and Training relevant to the Project

Expert in applying computational biology tools, including Neural network, HMM-based learning algorithm, molecular docking, in-silico compound library screening and molecular dynamics simulations, to solve problems in protein-ligand interaction studies.

B. Peer-reviewed publications (Publications in chronological order)

1. Ivanna Ihnatovych, **Tapan K Nayak**, Aya Ouf, Norbert Sule, Barbara Birkaya, and Anthony Auerbach, *Kinga Szigeti. (2019). iPSC model of CHRFAM7A effect on $\alpha 7$ nicotinic acetylcholine receptor function in the human context. *Translational Psychiatry*. 9(1), 59.

2. **Tapan K Nayak**, Viz, R, Bruhova, I, Shandilya, J, *Auerbach A. (2019). Efficiency measures the conversion of agonist binding energy into receptor conformational change. *J. Gen. Physiol.*, 151(4), 465.
3. **Tapan K Nayak**, *Auerbach A. (2017). Cyclic activation of endplate acetylcholine receptors. *Proc. Natl. Acad. Sci. U.S.A.*, 114(45):11914-11919.
4. **Tapan K Nayak**, Chakrabarty S, Zheng W, *Auerbach A. (2016). Structural correlates of affinity in fetal versus adult endplate nicotinic receptors. *Nature Communications* 7:11352; DOI: 10.1038/ncomms11352.
5. Shandilya J, Gao Y, **Tapan K Nayak**, Roberts S, *Medler K. (2016). AP1 transcription factors are required to maintain the peripheral taste system. *Cell Death and Disease*. 7(10):e2433. DOI: 10.1038/cddis.2016.343.
6. **Tapan K Nayak**, Bruhova I, Chakrabarty S, Gupta S, Zheng W, *Auerbach A (2014). Functional differences between neurotransmitter binding sites of muscle acetylcholine receptors. *Proc. Natl. Acad. Sci. U.S.A.*, 111(49): 17660-05.
7. Iacobucci G, Rahman NA, Valtuena AA, **Tapan K Nayak**, *Gunawardena S (2014). Spatial and temporal characteristics of normal and perturbed axonal transport *in vivo*. *PLoS One*, 9(5):e97237.
8. **Tapan K Nayak**, *Auerbach A (2013). Asymmetric transmitter binding sites of fetal muscle acetylcholine receptors shape their synaptic response. *Proc. Natl. Acad. Sci. U.S.A.*, 110(33): 13654-59.
9. **Tapan K Nayak**, Purohit PG, *Auerbach A (2012). The intrinsic energy of the gating isomerization of a neuromuscular acetylcholine receptor channel. *J. Gen. Physiol.*, 139(5):349-58.
10. Tapan K Nayak, Dana S, Raha S, Sikdar SK (2011). Activator-induced dynamic disorder and molecular memory in human two-pore domain hTREK1 K⁺ channel. *J. Chem. Biol.* 4(2):69-84.
11. **Tapan K Nayak**, Harinath S, Nama S, Somasundaram K, *Sikdar SK (2009). Inhibition of human two-pore domain K⁺ channel TREK1 by local anesthetic lidocaine: negative cooperativity and half-of-sites saturation kinetics. *Molecular Pharmacology*. 76(4): 903-17.
12. **Tapan K Nayak**, Sikdar SK (2007). Time-dependent molecular memory in single voltage-gated sodium channel. *J. Memb. Biol.* 219(1-3): 19-36.

Book Chapter

Nadira Khatoon, Adusumilli S, Dey P, Sharma R, Kampani P, Shandilya J, **Tapan K Nayak** (2022). Protein engineering and design in ion channels and receptors. *Methods in Cell Biology*. 169: 143-168.

Under review

1. Monika Kumari, Khatoon N, Adusumilli S, Sharma R, Auerbach A, Kashyap H, Tapan K Nayak. (2022). Hydrophobic gating of the acetylcholine receptor channel pore. *Under Revision (J. Gen. Physiology)*.